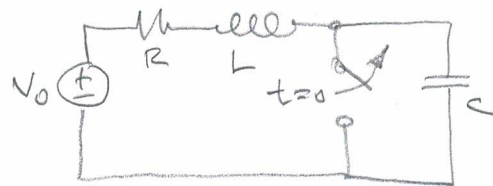


5.76)



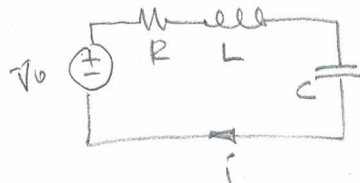
$$R = 4 \Omega \quad V_0 = 12 \text{ V}$$

$$L = 1 \text{ H}$$

$$C = \frac{1}{2} \text{ F}$$

Bestimme Maximalwert für V_C für $t \geq 0$.

Lösung: für $t \geq 0$ gilt



KSPL gem

$$-V_0 + Ri + V_L + V_C = 0 \quad (1)$$

Dfr

$$\begin{cases} V_L = L \frac{di}{dt} \\ V_C = V_C(0) + \frac{1}{C} \int i dt \end{cases} \quad (2)$$

(2) i(t) gem

$$-V_0 + Ri + L \frac{di}{dt} + V_C(0) + \frac{1}{C} \int i dt = 0 \quad (3)$$

Deriviere nach t

$$R \frac{di}{dt} + L \frac{d^2 i}{dt^2} + \frac{1}{C} i = 0$$

$$\frac{d^2 i}{dt^2} + \underbrace{\frac{R}{L}}_{2\alpha} \frac{di}{dt} + \underbrace{\frac{1}{LC}}_{\omega_0^2} i = 0$$

$$\Rightarrow \begin{cases} \alpha = \frac{R}{2L} = \frac{4}{2 \cdot 1} = 2 \\ \omega_0 = \frac{1}{\sqrt{LC}} = 2 \end{cases}$$

$\alpha = \omega_0$
kritisch gedämpft
System.

Lösungsweg Burke

$$i(t) = A e^{-\alpha t} + B t e^{-\alpha t} \quad (4)$$

für $t=0$ gilt es an $i(0) = \frac{V_0}{R}$ dispoen für

"Stromstärke"

$$(4) \Rightarrow A = \frac{V_0}{R} \quad (5)$$

(3) verwenden für $t=0$

$$(3) \Rightarrow -V_0 + Ri(0) + L \frac{di}{dt} = 0$$

$$(5) i(3) \text{ gem } -V_0 + Ri(0) + L \left(-A\alpha e^{-\alpha t} + B e^{-\alpha t} - B t e^{-\alpha t} \right) \Big|_{t=0} = 0$$

$$\Rightarrow -V_0 + R \frac{V_0}{R} + L \left(-\frac{V_0}{R} \alpha + B \right) = 0 \quad \hookrightarrow$$

$$-\frac{V_0}{R} \alpha = -B \Rightarrow B = \frac{V_0}{R} \alpha = \frac{12}{4} 2 = \underline{6}$$

Lösningen för strömmen blir alltså

$$\underline{i(t) = 3e^{-2t} + 6te^{-2t} \text{ A.}}$$

MEN DET VÄRKN SPÄNNINGEN ÖVER KOND OM FÖRSTÄDES
ANVÄND (3)

$$-V_0 + Ri + L \frac{di}{dt} = -V_C$$

VI FÅR

$$\begin{aligned} V_0 - 4(3e^{-2t} + 6te^{-2t}) - 1(-6e^{-2t} + 6e^{-2t} - \\ - 12te^{-2t}) = \underline{12 - 12e^{-2t} - 12te^{-2t} = V_C} \end{aligned}$$

SVAR: $V_C(t) = 12 - 12e^{-2t} - 12te^{-2t} \text{ VOLT}$